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Name	:	**Zahoor**	**Lab Ref number**	:	MRD-07/2024
Age/Sex	:	37 Y Male	**Receiving date**	:	10th January, 2025
MAX-ID No	:	SKDD.986664	**Reporting date**	:	10th January, 2025
Consultant	:	Self	**OPD/ IPD**	:	OPD

Flow Cytometric Immunophenotyping Report MRD-07/2025

Clinical History:	**Known case of AML for MRD evaluation.**
Specimen type:	Bone Marrow in EDTA
Specimen Quality:	Partially hemodilute. Few PDCs and basophils and a very few plasma cells were identified. No mast cells, hematogones are seen.
Specimen Age:	Fresh sample (<24 hours old).
CD Markers used	CD38, CD19,CD45,CD36,CD64,CD13,CD33,CD123,CD117,HLADR,CD34,CD7,CD15, CD11b, CD56
Instrument used	BD FACS Lyric
Software	BD FACS Suite Software
Lower limit of detection (LLOD)	0.001%
Lower limit of quantification (LLOQ)	0.003%
Diagnostic LAIP (If available)	Not Available.
Descriptive summary	12-colour flow-cytometry: Gating done using forward-side scatter, CD45-side scatter and CD34-side scatter analysis, CD34/CD117 analysis, and Boolean gating strategy to assess residual blast cells. ~1.2 million events were acquired with > 100 events in the blast gate. Both different from normal (DfN) and Leukemia associated aberrant immunophenotype (LAIP) approach used for MRD calculation.
Total Cells acquired	1,233,054
Viable cells	651,805
Events defining MRD	**0**
% MRD	**0**
MRD phenotype:	**The abnormal myeloid blasts are dim for CD45 with moderate side scatter (blast window). The blasts express bright CD34, moderate CD117, moderate CD13, CD38, dim-negative CD33, CD123, heterogenous CD64, bright HLA-DR, and negative for CD7, CD14, CD15, CD11b, CD36 and CD19.**
Normal Myeloid blasts	0.1% of all viable cells
Hematopoietic stem cells (HSCs)	0.02% of all viable cells.
Conclusion	**Multi-parametric flow cytometric MRD of the partially hemodilute sample shows 0.23% residual myeloid blast cells.**

Comment: *The clinically relevant actionable cut-off for MRD positivity for AML is 0.1%*. LSCs require close follow-up.*

****Reference***: *Schuurhuis GJ, Heuser M, Freeman S, Béné MC, Buccisano F, Cloos J, Grimwade D, Haferlach T, Hills RK, Hourigan CS, Jorgensen JL. Minimal/measurable residual disease in AML: a consensus document from the European LeukemiaNet MRD Working Party. Blood. 2018 Mar 22;131(12):1275-91.*

Note: Reporting format as per ELN guidelines 2018 (Ref. Blood. 2018;131(12):1275-1291)

Reported by:
Dr. Mousumi Kar
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Dated:- 10th January, 2025

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